

Clinical Investigation: Central Nervous System Tumor

Outcomes of Proton Radiation Therapy for Peripapillary Choroidal Melanoma at the BC Cancer Agency

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Received May 2, 2011, and in revised form Sep 26, 2011. Accepted for publication Oct 4, 2011

Summary

We report our experience with proton beam therapy in the treatment of peripapillary choroidal melanomas by means of a retrospective analysis of 59 patients treated between 1995 and 2007 at the only Canadian proton therapy facility. This article shows that excellent control and toxicity rates can also be achieved at smaller proton centers, which is of importance as newer centers come online and may provide this treatment option for localized choroidal melanomas. These results also raise caution for larger (T3) peripapillary tumors, because this group appears to have worse local control and distant metastasis rates.

Purpose: To report toxicity, local control, enucleation, and survival rates for patients with peripapillary choroidal melanoma treated with proton therapy in Canada.

Methods and Materials: We performed a retrospective analysis of patients with peripapillary choroidal melanoma (≤ 2 mm from optic disc) treated between 1995 and 2007 at the only Canadian proton therapy facility. A prospective database was updated for follow-up information from a chart review. Descriptive and actuarial data are presented.

Results: In total, 59 patients were treated. The median age was 59 years. According to the 2010 American Joint Committee on Cancer TNM classification, there were 20 T1 tumors (34%), 28 T2 tumors (48%), and 11 T3 tumors (19%). The median tumor diameter was 11.4 mm, and the median thickness was 3.5 mm. Median follow-up was 63 months. Nineteen patients received 54 cobalt gray equivalents (CGE) and forty patients received 60 CGE, each in 4 fractions. The 5-year actuarial local control rate was 91% (T1, 100%; T2, 93%; and T3, 59%) ($p = 0.038$). There was a suggestive relationship between local control and dose. The local control rate was 97% with 60 CGE and 83% with 54 CGE ($p = 0.106$). The metastasis-free survival rate was 82% and related to T stage (T1, 94%; T2, 84%; and T3, 47%) ($p < 0.001$). Twelve patients died, including eleven with metastases. The 5-year actuarial rate of neovascular glaucoma was 31% (23% for T1–T2 and 68% for T3, $p < 0.001$), and that of enucleation was 0% for T1, 14% for T2, and 72% for T3 ($p < 0.001$). Radiation retinopathy (74%) and optic neuropathy (64%) were common within-field effects.

Conclusions: Proton therapy provides excellent local control with acceptable toxicity while conserving the globe in 80% of cases. These results are consistent with other single-institution series using proton radiotherapy, and toxicity rates were acceptable. T3 tumors carry a higher rate of both local recurrence and metastasis. © 2012 Elsevier Inc.

Keywords: Proton therapy, Choroidal melanoma, Peripapillary, Local control, Survival

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Presented in part at the annual scientific meeting of the Canadian Association of Radiation Oncology, Vancouver, British Columbia, Canada, September 22–25, 2010.

Conflict of interest: none.

Acknowledgment—The authors thank Drs. William Kwa and Cheryl Duzenli, medical physicists, for their valuable contributions to the proton therapy program and Dr. Robert Olson, radiation oncologist, for assistance with statistical analyses.

Introduction

Choroidal melanoma is a relatively uncommon malignancy, with an incidence rate of 4.3 per million each year in the United States (1). After the results of the large randomized Collaborative Ocular Melanoma Study (COMS) trial, eye-sparing treatments such as episcleral plaque brachytherapy, proton beam treatment, and stereotactic radiotherapy (SRT) have largely replaced enucleation in the treatment of localized choroidal melanomas (2–4). The principal advantage of radiotherapy is eye conservation and retention of useful vision in many patients, without sacrificing local control or survival.

Brachytherapy does not achieve adequate coverage of peripapillary choroidal melanomas because a tumor located next to the optic nerve hinders placement of the temporary plaque onto the sclera. Placement of a radioactive plaque adjacent to the optic nerve results in an excessively high risk of long-term sequelae including blindness (5).

Several strategies have been used to circumvent these limitations, including notched plaque radiotherapy, charged particle radiotherapy, and stereotactic radiosurgery (3, 4, 6–9). The dosimetric characteristics of protons allow for a relatively uniform dose to be delivered to the entire tumor, as well as a sharp reduction in dose outside the treated area, as compared with high-energy photon radiotherapy.

Since August 1995, the BC Cancer Agency, in conjunction with the University of British Columbia Eye Care Centre and TRIUMF (TRI-University Meson Facility), has been operating the only proton therapy facility in Canada to treat patients with ocular melanomas. Indications for proton therapy include large tumors and peripapillary tumors unsuitable for brachytherapy. This is the first report of toxicity, local control, and survival rates for patients with peripapillary choroidal melanomas treated with proton therapy in Canada.

Methods and Materials

Patients were referred to the ocular oncology clinic at the Eye Care Centre in Vancouver by ophthalmologists based primarily in Western Canada. Demographic and tumor data were prospectively recorded into a database. Patients with peripapillary choroidal melanoma, defined as less than or equal to 2 mm from the optic disc, treated between March 1995 and December 2007 with proton therapy were retrospectively selected for this study. Patients with nonmelanoma tumors or re-treatment of plaque failures were excluded.

Before proton treatment, all patients had full systematic ocular examinations by a dedicated ocular oncologist (K.P.). This included slit-lamp examination, visual acuity, intraocular pressure measurement, direct and indirect ophthalmoscopy, fundus photography, fluorescein angiography, and A- and B-mode ultrasound for tumor measurements. Staging investigations included chest radiography, liver enzyme levels, and liver ultrasounds to exclude overt metastases. The patients were also assessed by one of two radiation oncologists (T.P. or R.M.) at the BC Cancer Agency to confirm eligibility.

Visual acuity and intraocular pressures were recorded prospectively. Slit-lamp examination and gonioscopy were performed to look for new iris and angle vessels at each visit. Tumor measurements were regularly recorded with ultrasonography and

funduscopy. Treatment-related complications were documented prospectively.

The prospective database was updated for this report based on review of charts at the ocular oncology clinic and the BC Cancer Agency. Follow-up information for out-of-province and international patients was obtained from referring ophthalmologists through written and/or telephone contact. Tumors were retrospectively classified into T stage according to the 2010 American Joint Committee on Cancer's TNM cancer staging manual, based on tumor thickness and diameter at diagnosis (10). This study was reviewed and approved by the Research Ethics Board of the University of British Columbia.

Treatment procedure

The proton therapy treatment procedure is similar to that described by other authors (3, 7). Up to six tantalum clips, each measuring 2 mm, were surgically implanted on the sclera around the tumor edges after localization by transillumination and indirect ophthalmoscopy. This procedure was done with the patient under general anesthesia approximately 1 week before the planned treatment date. The distance between each clip and tumor edge was recorded intraoperatively.

The treatment was simulated by use of a thermoplastic head shell and bite block for immobilization (Fig. 1). Once installed on the positioning chair, the patient was asked to fixate on a small flashing target. Clip positions were captured by use of orthogonal x-ray images. Treatment was planned by use of EYEPLAN software, developed originally by Goitein and Miller (11) for dosimetry based on three-dimensional reconstruction of the eye, ocular structures, and tumor. Tumor borders were delineated by the ocular oncologist and verified by the radiation oncologist. Various gaze angles were assessed by the ophthalmologist and medical physicist to select the optimal angle that minimized irradiation of normal tissues. A brass collimator was custom-made for each patient to shape the treatment beam with a 2.5-mm margin to beam edge.

Orthogonal films were taken before each treatment to verify positioning of the clips relative to the treatment plan, and patient position adjustments were made if necessary. Treatment was delivered in 4 fractions on 4 consecutive days at the TRIUMF cyclotron facility on the University of British Columbia campus. T stage was not a determinant in dose; instead, the dose was initially 60 cobalt gray equivalents (CGE) for all patients until 1997. The dose was reduced to 54 CGE from 1997 to 2001 because of a clinical impression of a high rate of secondary glaucoma. The dose was reverted back to 60 CGE from 2001 onward because of the clinical concern of loss of tumor control observed on follow-up. The proton beam energy was set at 74 MeV, with a practical range in tissue of 34.3 mm, defined by the 90% isodose curve. A 2.5-mm margin was used for the distal and proximal end of the defined target. The beam-on time for each treatment session was approximately 90 seconds. The eye position was closely monitored for movement while the radiation beam was on, and the treatment was interrupted if necessary. After completion of the treatment, the patient was regularly evaluated for tumor response, toxicity, and visual acuity by the ocular oncologist.

Statistical analyses

Estimates of local control, metastasis-free survival, and overall survival were calculated according to the actuarial method of

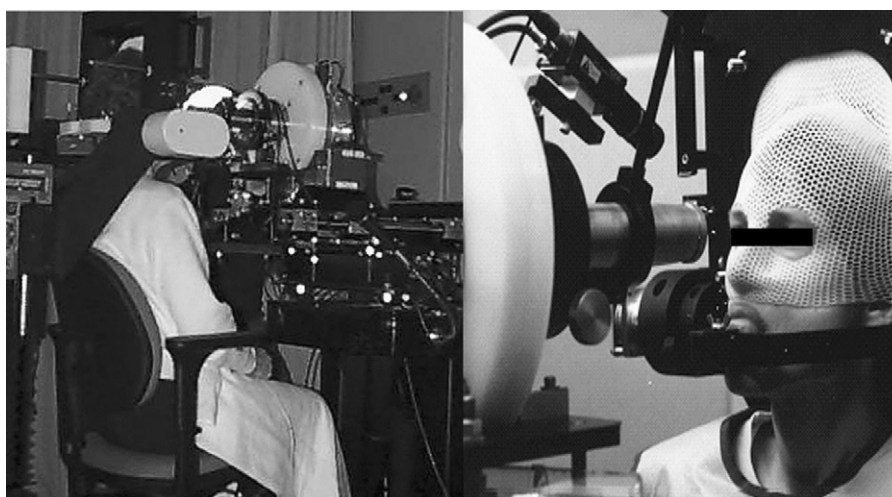


Fig. 1. Patient immobilized in treatment position by use of thermoplastic head shell and bite block.

Kaplan and Meier. Rates of secondary enucleation and radiation complications were calculated by use of the cumulative incidence approach at 2 and 5 years. Outcomes were measured from the start date of proton radiotherapy to the date of event or last follow-up. Comparisons for various endpoints were performed by use of the log-rank (Mantel-Cox) test to identify important prognostic factors. Intervals of significance were represented by *p* values derived from two-sided tests. Statistical analyses were performed with SPSS software, version 14.0 (SPSS, Chicago, IL).

Results

A total of 123 patients were treated with proton therapy between 1995 and 2007. We excluded 60 patients, comprising 4 with large iris melanomas, 5 with recurrent disease after eye-conserving treatment, 21 with large anterior choroidal melanomas, and 30 with posterior choroidal melanomas, because their tumors were more than 2 mm from the optic disc. Among 63 patients with peripapillary tumors treated during the study period, 59 met the eligibility criteria and are the subjects of this report. Two patients with nonmelanoma tumors (hemangiomas) and two re-treatments of plaque failure were excluded. Patient demographics are summarized in Table 1. The median age was 59 years, with a range from 34 to 86 years. There were 39 men and 20 women. Of the patients, 32 lived in British Columbia, 25 lived elsewhere in Canada, and 2 lived out of the country. No patients had metastatic disease at the time of treatment. Medical history included hypertension in 23 patients (39%), diabetes in 7 (12%), and cerebrovascular disease in 2 (3%). Two patients had pre-existing open-angle glaucoma.

There were 20 T1 tumors (34%), 28 T2 tumors (47%), and 11 T3 tumors (19%). The median tumor thickness was 3.5 mm, and the median maximal tumor base diameter was 11.4 mm. None of the patients had ciliary body or iris involvement. One patient had possible extrascleral extension at diagnosis (prominent optic nerve on magnetic resonance imaging). Nineteen patients were treated with 54 CGE in 4 fractions, whereas the remaining forty patients received 60 CGE in 4 fractions. All patients received their prescribed proton treatments without a significant treatment break. The median treatment depth used was 24.7 mm (range,

17.8–28.6 mm). The median follow-up was 63 months, with a range from 4 to 131 months. The median follow-up was 105 months and 48 months for patients receiving 54 CGE and those receiving 60 CGE, respectively. Five patients did not have complete follow-up information (2 lived out of the province and 1 lived out of the country), with a median follow-up of 42 months for these patients (range, 4–107 months).

Table 1 Demographics and tumor and treatment characteristics of 59 patients with choroidal melanoma treated with proton therapy (1995–2007)

Parameter	Data
Age (yr)	
Median	59
Range	34–86
Gender [n (%)]	
Male	39 (66)
Female	20 (34)
Laterality [n (%)]	
Left	32 (54)
Right	27 (46)
Tumor characteristics	
Median thickness (mm)	3.5 (range, 1.5–11.5)
Median diameter (mm)	11.4 (range, 6.8–17)
Median distance to OD (mm)	0 (range, 0–2)
T stage* [n (%)]	
T1	20 (34)
T2	28 (47)
T3	11 (19)
Dose [n (%)]	
54 CGE	19 (32)
60 CGE	40 (68)
Pretreatment ipsilateral visual acuity [n (%)]	
≥20/50	31 (53)
≥20/200	45 (76)
<20/200	14 (24)

Abbreviations: OD = optic disc; CGE = cobalt gray equivalent.

* 2010 American Joint Committee on Cancer clinical T stage.

Outcomes are summarized in Table 2. The 5-year local control rate was 91%. Five local recurrences were detected during follow-up. The median interval between radiation treatment and local recurrence for these 5 patients was 24 months (range, 12–97 months). Management for local recurrence was enucleation in 4 patients and transpupillary thermotherapy without enucleation in 1. After local recurrence, metastases developed in 2 subjects and they died of disease. Actuarial local control rates at 5 years for T1, T2, and T3 tumors were 100%, 93%, and 59%, respectively ($p = 0.038$) (Fig. 2). The difference in local control for patients receiving 60 CGE and those given 54 CGE was not significant (97% and 83%, respectively, at 5 years; $p = 0.106$) (Fig. 3). No other factors were associated with local control on univariate analysis.

The 5-year metastasis-free survival rate was 82%. Metastases were detected in 15 patients, with a median interval from irradiation to metastasis of 38 months (range, 11–123 months). Actuarial metastasis-free survival rates at 5 years for T1, T2, and T3 tumors were 94%, 84%, and 47%, respectively ($p < 0.001$) (Fig. 4). The 5-year overall survival rate was 85%. Age at diagnosis and gender were not associated with increased risk of death from melanoma. Twelve deaths occurred during the follow-up period, and the cause of death was metastatic melanoma in eleven patients and a cerebrovascular event in one.

Neovascular glaucoma was observed in 17 patients (31%) followed up for 5 years. The majority of affected patients (67%) were successfully treated with medical therapy. There was no relationship between prescribed dose and occurrence of

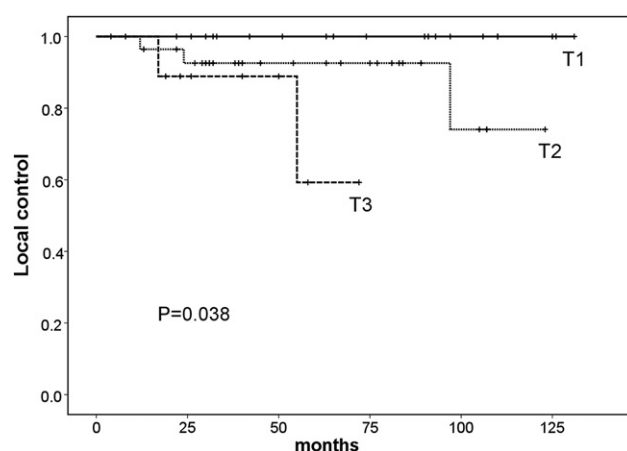


Fig. 2. Kaplan-Meier curve showing local control rates by T stage.

neovascular glaucoma on univariate analysis ($p = 0.986$, log-rank test). Neovascular glaucoma developed at 17 months after irradiation in 1 of the 2 patients with pre-existing glaucoma, and neither required enucleation. Five patients required enucleation for refractory pain. Intravitreal bevacizumab was effectively used in 6 patients to avoid enucleation (12). Actuarial 5-year rates of radiation retinopathy, optic neuropathy, vascular occlusion, and cataract formation were 74%, 64%, 19%, and 30%, respectively.

Enucleation rates were 0% for T1 tumors, 14% for T2 tumors, and 72% for T3 tumors ($p < 0.001$), and the overall eye-conservation rate was 80%. Enucleation was required in 12 patients because of local recurrence ($n = 4$), painful neovascular glaucoma ($n = 5$), or complete opacification of the ocular lens rendering tumor assessment unreliable ($n = 2$) or for an unclear reason ($n = 1$).

Seventy-six percent of patients had a best corrected pretreatment visual acuity of better than or equal to 20/200 in the affected eye. Among 31 patients with visual acuity recorded at 5 years after treatment, 17% maintained a best corrected visual acuity of better than or equal to 20/200, whereas 42% would be able to count fingers.

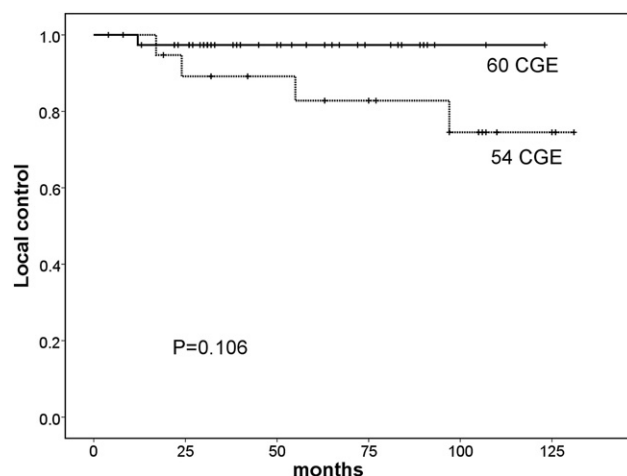


Fig. 3. Kaplan-Meier curve showing local control rates by prescribed dose. CGE = cobalt gray equivalent.

Table 2 Treatment outcome for 59 patients with choroidal melanoma treated with proton therapy (1995–2007)

Outcome	2 yr (%)	5 yr (%)
Local control		
Overall	94.5	91.3
T1	100	100
T2	92.6	92.6
T3	88.9	59.3
Overall survival	96.5	85.0
Metastasis-free survival		
Overall	91.2	82.1
T1	100	94.4
T2	89.3	83.7
T3	77.8	46.7
Enucleation rate		
Overall	8.6	20.0
T1	0	0
T2	3.7	13.5
T3	36.4	71.7
Neovascular glaucoma		
Overall	19.3	31.1
T1	10.0	23.4
T2	14.8	23.9
T3	52.3	68.2
Radiation retinopathy	58.0	73.5
Optic neuropathy	45.4	63.6
Vascular occlusion	8.7	19.1
Cataract formation	16.2	29.6
Useful vision ($\geq 20/200$)	41.2	17.0
Count fingers ($\geq 20/1,000$)	65.0	59.2

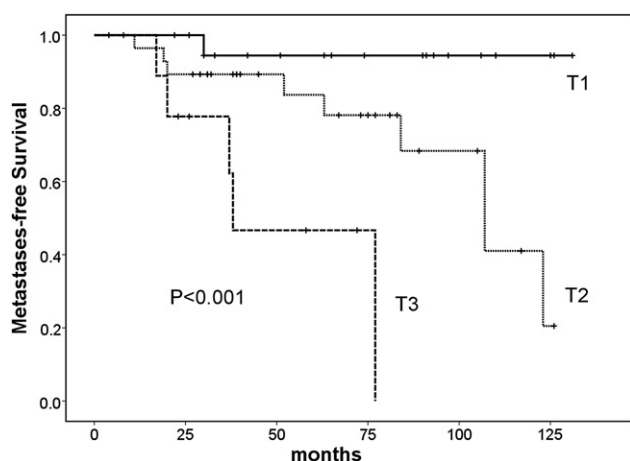


Fig. 4. Kaplan-Meier curve showing metastasis-free survival rates by T stage.

Discussion

Proton therapy at the BC Cancer Agency is reserved for patients with tumors that are not suitable for brachytherapy, including large and peripapillary tumors. The advantages of proton therapy include a high degree of conformity to the target volume and a steep dose gradient, which allows superior sparing of normal tissues as compared with other techniques. A study comparing proton therapy, Gamma Knife (Elekta, Stockholm, Sweden), and CyberKnife (Accuray, Sunnyvale, CA) using an anthropomorphic phantom showed that peripheral doses to the contralateral eye and thyroid were lowest with proton therapy (13).

To date, excellent local control rates (90.5–99%) with proton therapy have been reported (3, 7–9). One of the largest series included 2,069 patients treated at Harvard Cyclotron Laboratory and Northeast Proton Therapy Center between 1975 and 1997 (3). That study reported local control and eye-conservation rates of 95% and 84%, respectively. The importance of achieving local control has been emphasized by some authors, because treatment failure has been linked to development of metastatic disease (8, 14).

The 5-year local control rate in our series was 91% with a median follow-up of 63 months. This is despite the fact that all tumors were peripapillary. Several retrospective studies have reported higher local failure rates when tumors adjacent to the optic nerve or macula were treated with plaque brachytherapy (15, 16). The landmark COMS medium-tumor trial excluded patients with peripapillary tumors because they were considered to have a higher risk of tumor recurrence and distant metastases (2). This is likely due to the inherent difficulty of securing the plaque because of the location of the optic nerve, leading to inferior local control outcomes. In addition, a randomized trial comparing helium-charged particles with iodine-125 plaque therapy showed that local control was significantly higher in the helium arm (100% vs. 87%) with lower enucleation rates (9). As such, proton therapy may be preferred for peripapillary tumors. Our series restricted patient selection for proton therapy to patients with peripapillary tumors, whereas other centers treat all referred choroidal melanomas with proton therapy, regardless of size. This may account for the local control rate of 99% observed in the series by Egger *et al.* (8).

The staging for choroidal melanoma was substantially revised in the seventh edition of the *AJCC Cancer Staging Manual*, based on a collaborative database of over 7,000 patients with choroidal melanoma (10). In our series using the new classification, the predictive value of accurately determining pretreatment T stage was confirmed. T stage was the single most useful prognostic factor to predict the risk of local relapse, distant metastases, and ocular complications. However, assessment of other risk factors may not be reliable because of the low incidence of relapses overall. T1 tumors had the lowest rates, and none required enucleation. By contrast, T3 tumors have a 5-year metastasis-free survival rate of 47%, and 7 patients required enucleation. Tumor size has consistently been shown to be of prognostic significance in terms of metastasis and death (17–19). Data from Gragoudas *et al.* showed that 65% of cases in which metastases develop after proton therapy had tumors with a base diameter larger than 15 mm (18). The outcome of metastatic choroidal melanoma remains very poor, with a median survival ranging from 2 to 12 months (20).

A dose response is suggested in this study, although this was not statistically significant on univariate analysis. There was only 1 local recurrence with a dose of 60 CGE, whereas 4 local recurrences were detected with a dose of 54 CGE. Although patients treated with 54 CGE had longer median follow-up (105 months vs. 48 months), the median time to recurrence is 24 months and most events would have been captured. No local recurrences were noted among the first 15 patients treated with proton therapy (the dose was 60 CGE for the first 10 patients), rendering the possibility of a treatment-era effect less likely. However, when the dose dropped to 54 CGE, 4 of the subsequent 19 patients had failure locally. The small number of patients and event rates would also make a multivariate analysis unreliable, and therefore such an analysis was not performed. Despite a suggestive relationship between local control and dose in this study, a randomized clinical trial of dose in proton therapy compared 70 CGE and 50 CGE (28% reduction) in 5 fractions in 188 patients with small- to medium-sized choroidal tumors (21). This study showed no difference in 5-year local control rates, survival, and ocular toxicity. However, tumors were slightly smaller in terms of both median thickness (3.0 mm) and diameter (10.0 mm), which may account for the study's low rate of tumor recurrences overall.

Subsequent enucleation was performed in 12 patients (20%) in this series. Most patients underwent enucleation because of intractable pain from neovascular glaucoma (42%) or local recurrence (33%). The observed enucleation rate was similar to rates reported by other authors (3, 7–9). The enucleation rate is expected to decrease as intravitreal bevacizumab, an antibody against angiogenesis, is introduced into the management of neovascular glaucoma. A single administration of intravitreal bevacizumab resulted in rapid regression of new iris and angle vessels and led to control of intraocular pressure and conservation of the globe in 2 of 3 patients with neovascular glaucoma (12).

Radiation retinopathy and optic neuropathy were common treatment-related late effects in this study. These are slowly progressive and unavoidable complications of proton therapy because the adjacent optic nerve and often the macula must be in the treatment field and have received a radiation dose that exceeds the tolerance of neural tissue. Visual outcome analysis was therefore predictably disappointing, with a rate of useful vision ($\geq 20/200$) of only 17% at 5 years in our cohort.

The poor outcomes of T3 tumors in our cohort is concerning, with 41% of cases failing locally and 53% failing distally at 5

years. Even when patients were subsequently treated by enucleation, tumor recurrences have consistently been reported to be associated with increased risk of metastases and death. The optimal selection of treatment for patients with T3 tumors is debatable, but there is some suggestion that these patients should undergo enucleation upfront (22). A potential alternative for patients wishing to avoid upfront enucleation is to dose escalate for T3 tumors to improve the tumor control rate and to use intravitreal bevacizumab to address the added toxicity from a higher dose to normal ocular structures. Another option is to combine hyperthermia, a known radiation sensitizer, with proton therapy to improve local control (23). A third strategy would be to better address marginal failures, which may be prevented by improvement in the planning system used to accurately model the tumor. Daftari *et al.* (24) studied 79 patients with peripapillary tumors planned with an earlier version of EYEPLAN treated with protons. Using an updated version of EYEPLAN, which allowed digital fundus photographs to be incorporated into the planning procedure, they were able to determine, on replanning, that 2 of 6 local failures had the target volume inadequately covered by the original plan. Their study showed that some local failures might have been prevented with better planning techniques. However, there are currently no prospective comparative trials evaluating these different options for large peripapillary choroidal melanomas.

Over the past decade, SRT has been used increasingly as an alternative for peripapillary choroidal melanomas in large part because of the greater availability of linear accelerators compared with proton treatment facilities. Multiple centers have evaluated the feasibility and efficacy of SRT in choroidal melanomas and reported local control outcomes comparable to proton therapy (4, 25). A practical advantage of SRT is that it can spare the patient the surgical insertion of localization clips. However, tumor delineation based on computed tomography images as used in SRT can be challenging and should be complemented by fusion with magnetic resonance imaging, digital fundus photography, and the use of intravenous contrast.

In the SRT series by Krema *et al.* (4), the rate of neovascular glaucoma was 42% with a median follow-up of just 37 months. This is higher than reported in the proton literature. The follow-up for most SRT series remains relatively short, as compared with the proton therapy experience in which 15-year data have already been published. As such, Krema *et al.* concluded that SRT provides comparable tumor control but at the expense of a long-term higher incidence of ocular complications, particularly to the anterior segment. Further studies are needed to determine whether these risks are indeed higher after SRT compared with proton therapy.

Finally, our study has several limitations, and some of these data have to be viewed with caution. The study suffers from the inherent limitation of any retrospective analysis, it has a limited number of patients, and quality-of-life outcomes were not evaluated. Patients were also selected based on tumor characteristics, patient preference, and physician judgment. Although there may be some selection bias as a result, treatment was delivered at a population-based level in Western Canada and there was little variation in the treatment procedures since the start of the proton therapy program to the time of writing this manuscript. Another important limitation is the small number of local recurrence events, and therefore more detailed statistical analyses were not performed.

Conclusion

Our results confirm that proton radiation therapy is an excellent treatment for patients with peripapillary choroidal melanoma. The treatment is generally well tolerated, and most patients are able to avoid enucleation. The outcomes are comparable to previously published reports and support the continued use of proton therapy. Large T3 peripapillary tumors represent a higher-risk subgroup with respect to local recurrence, distant metastases, and ocular complications. Further studies are warranted to investigate the optimal selection of treatment in these patients.

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